CMPE 258 Project Proposal

Graph-Based Intelligent Movie Recommendation System Using RAG

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A. Proposed Application Area

Option 2 — LLMs + AI Agent System (Evaluation-First)

B. Problem Definition

Traditional movie recommender systems primarily rely on collaborative filtering or content-based filtering. These approaches often suffer from cold-start problems, popularity bias, and limited explainability. We propose a hybrid Knowledge Graph + Retrieval-Augmented Generation (RAG) framework that retrieves structured relationships from a movie knowledge graph and combines them with semantic embeddings to generate context-aware and explainable recommendations using LLMs. Our goal is to improve recommendation relevance, interpretability, and robustness while conducting a rigorous evaluation of different LLM-based retrieval architectures.

C. Identified Dataset and Features

We use the Neo4j "movie recommendations" dataset, which contains approximately 28,865 nodes and 166,262 relationships organized in a rich relational graph structure. The primary node types include Movie, Genre, User, Actor, Director, and Person.

Key relationships include (:Movie)-[:IN_GENRE]->(:Genre), (:Actor)-[:ACTED_IN]->(:Movie), (:Director)-[:DIRECTED]->(:Movie), and (:User)-[:RATED {rating}]->(:Movie).

The dataset provides movie metadata (title, release year, genre), user rating scores, and actor/director associations. These structured graph relationships enable efficient traversal and filtering using Cypher queries, while also supporting semantic retrieval when converted into knowledge triplets and embedded into vector space. This combination allows grounded, explainable recommendation generation through hybrid graph-based and neural retrieval methods.

D. Final Demo

The final demo will present a web-based Streamlit application where users can submit natural language movie recommendation queries and receive grounded, explainable responses. The system will implement a hybrid GraphRAG architecture that combines structured Neo4j Cypher queries for constraint-based filtering (e.g., genre, rating, release year, actor) with FAISS-based semantic retrieval for similarity reasoning. Retrieved knowledge will be passed to multiple LLMs to generate responses supported by explicit evidence from the Knowledge Graph. The interface will display retrieved supporting facts, model outputs from at least three LLMs (including one open-source model), and structured logs of retrieval steps and tool usage. System performance will be analyzed by comparing model outputs and reporting relevant metrics such as response quality and latency. Basic session handling and guardrails will be implemented to ensure constraint-respecting and grounded recommendations.

E. References

[1] P. Lewis *et al.*, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," *arXiv.org*, Apr. 12, 2021. https://arxiv.org/abs/2005.11401

[2] "KG-Retriever: Efficient Knowledge Indexing for Retrieval-Augmented Large Language Models," *Arxiv.org*, 2023. https://arxiv.org/html/2412.05547v1 (accessed Mar. 13, 2025).

[3] Z. Xu *et al.*, "Retrieval-Augmented Generation with Knowledge Graphs for Customer Service Question An-swering," 2024, doi: https://doi.org/10.1145/3626772.3661370.

[4] "Example datasets - Getting Started," *Neo4j Graph Data Platform*. https://neo4j.com/docs/getting-started/appendix/example-data/